

1 Bonus Optimization Experiment: Improving the LwF ImageNet $\rightarrow$ Scenes Results

As part of the bonus requirement, we performed an additional optimization experiment in order to improve the reproduced results of the original **Learning without Forgetting (LwF)** paper. The optimization was tested on the **ImageNet $\rightarrow$ MIT Indoor Scenes** continual learning scenario.

The goal was not to change the dataset or the evaluation protocol, but to improve the model behavior in a controlled and meaningful way. Therefore, the same MIT Indoor Scenes dataset, the same ImageNet validation subset, the same LwF training framework, the same temperature value, the same loss balance, and the same evaluation protocol were preserved. The only change was a targeted modification to the new-task classification head and its learning rate.

1.1 Motivation for the Optimization

In the standard LwF method, the model learns the new task while preserving the old task by combining two losses:

$$L_{\text{total}} = L_{\text{new}} + \lambda L_{\text{old}}$$

where L_{new} is the classification loss for the new MIT Indoor Scenes task, and L_{old} is the knowledge-distillation loss that keeps the ImageNet output behavior close to the original teacher model.

This structure is effective because it directly addresses the main problem in continual learning: learning a new task without causing catastrophic forgetting on the old task. However, in our reproduction, the standard LwF classifier still had room for improvement on the new Scenes task. Therefore, we added a more suitable classification head for the new task.

1.2 Optimization Method

The optimization used in the final version replaces the regular fully connected classifier of the new Scenes head with a **scaled cosine-normalized classifier**.

In a regular linear classifier, the class score is computed using a dot product:

$$z = f \cdot w$$

where f is the feature vector and w is the class weight vector. This dot product depends on both the direction and the magnitude of the vectors. In transfer learning and continual learning, this can be problematic because the classifier may become influenced by weight magnitudes instead of focusing only on the semantic similarity between the image representation and the class.

The cosine-normalized classifier solves this by normalizing both the feature vector and the class weights:

$$z = \eta \cdot \cos(f, w)$$

or explicitly:

$$z = \eta \cdot \frac{f \cdot w}{\|f\| \|w\|}$$

This means that each class is compared mainly according to angular similarity. In other words, the classifier focuses on how well the image representation matches the class direction, rather than being affected by uncontrolled growth in feature or weight norms.

This is especially useful in our experiment because the shared AlexNet representation was originally trained for ImageNet, while the new task is MIT Indoor Scenes. A cosine classifier gives the new task head a cleaner and more balanced decision rule.

1.3 Learnable Scale Parameter

Because cosine similarity is naturally limited to the range $[-1, 1]$, the classifier also uses a learnable scale parameter η . Without this scale, the logits may be too small, producing a softmax distribution that is too flat and weak gradients during training.

In the final optimized version, the initial scale was set to:

$$\eta_0 = 30$$

This value was chosen because a smaller scale was not strong enough for the Scenes classifier. With $\eta_0 = 30$, the classifier produced sharper and more useful logits for multi-class classification.

During training, η remained learnable, and the final run reached:

$$\eta = 30.26$$

This shows that the initial value was stable and appropriate, since the model did not need to move far away from it during optimization.

1.4 Boosted Learning Rate for the New Task Head

A second important part of the optimization was increasing the learning rate only for the new Scenes head:

$$LR_{\text{new head}} = 5 \times LR_{\text{base}}$$

The shared backbone and the old ImageNet head continued to train with the regular base learning rate. This is a strong design choice because it increases the flexibility of the new classifier without aggressively changing the shared representation.

In continual learning terms, this improves the balance between:

- **Plasticity** – the ability to learn the new Scenes task.
- **Stability** – the ability to preserve the old ImageNet task.

The new head needs to adapt quickly because it starts from random initialization and must learn 67 new scene classes. On the other hand, the shared layers already contain useful

visual features from ImageNet, so changing them too aggressively could damage the old task. Therefore, applying a higher learning rate only to the new head is a focused and controlled optimization.

1.5 Development of the Final Optimization

The optimization was developed in two stages.

First, we tested a cosine-normalized classifier with a lower scale value:

$$\eta_0 = 16$$

This version improved the old ImageNet task compared to standard LwF, but it did not improve the new Scenes task. The result showed that cosine normalization was useful for preserving stable representations, but the new task head was not learning strongly enough.

Based on this observation, the final version increased the initial scale to:

$$\eta_0 = 30$$

and also used a boosted learning rate for the new task head:

$$LR_{\text{new head}} = 5 \times LR_{\text{base}}$$

This final version improved the new task accuracy while still preserving the old task accuracy.

1.6 Results

The following table compares the standard LwF method, the first cosine classifier experiment, and the final optimized method.

Table 1: Comparison of LwF optimization stages for ImageNet $\rightarrow$ Scenes

Method	ImageNet Old Task Accuracy	Scenes New Task Accuracy
Standard LwF	59.82%	56.27%
LwF + Cosine Classifier, $\eta_0 = 16$	60.74%	52.31%
LwF + Cosine Classifier, $\eta_0 = 30$, Head LR $\times 5$	60.72%	58.06%

The final optimized method achieved:

$$58.06\%$$

on the MIT Indoor Scenes task, compared to:

$$56.27\%$$

for standard LwF.

Therefore, the new task improvement was:

$$58.06\% - 56.27\% = 1.79\%$$

The ImageNet old task accuracy also improved:

$$60.72\% - 59.82\% = 0.90\%$$

This is an important result because the optimization did not improve the new task by sacrificing the old task. Instead, it improved the new task accuracy while also slightly improving the old task preservation.

1.7 Comparison with Other Methods

The advantage of the optimized LwF method becomes clearer when comparing it with the other reproduced methods.

Table 2: Comparison between reproduced methods and the optimized LwF method

Method	ImageNet Old Task Accuracy	Scenes New Task Accuracy
Fine-Tuning	48.69%	50.82%
Feature Extraction	62.89%	55.00%
Standard LwF	59.82%	56.27%
Optimized LwF	60.72%	58.06%

Fine-tuning achieved only 50.82% on the Scenes task and reduced ImageNet accuracy to 48.69%, showing clear catastrophic forgetting. Feature extraction preserved ImageNet better, but reached only 55.00% on Scenes because the shared representation was not adapted enough to the new task.

The optimized LwF method provides a better balance. It achieved the highest Scenes accuracy among the reproduced methods while still preserving strong ImageNet performance.

1.8 Why This Optimization Works

This optimization is effective because it improves the most important part of the new-task learning process: the classifier that maps the shared visual representation into the 67 MIT Indoor Scenes classes.

The cosine-normalized classifier improves the decision boundary by making class scores depend on angular similarity instead of raw vector magnitudes. This makes the classifier more balanced and better suited for transfer learning.

The learnable scale parameter η makes the cosine logits strong enough for softmax classification. Without this scale, the classifier would produce weak class confidence and slower learning.

The boosted learning rate for the new task head allows the new classifier to adapt faster, while the shared layers remain protected by the regular LwF training process and the knowledge-distillation loss. This keeps the model stable on ImageNet while improving its ability to learn Scenes.

Overall, the optimization is successful because it improves the new task classifier without disrupting the core LwF mechanism. It keeps the main advantage of LwF: learning a new task without access to the old task training data, while preserving old task performance.

1.9 Conclusion

The final optimized method, **LwF with a scaled cosine classifier and boosted new-head learning rate**, improved the reproduced ImageNet $\rightarrow$ Scenes results.

Compared to standard LwF, it improved the MIT Indoor Scenes accuracy by:

$$+1.79\%$$

and improved the ImageNet old-task accuracy by:

$$+0.90\%$$

This makes the optimization meaningful because it improves the new task while maintaining old-task knowledge. The method is also efficient and controlled, since it does not require additional data, does not change the evaluation protocol, and does not require a larger model backbone. The improvement comes from a focused architectural and training modification that makes the new task head more suitable for continual learning.